

# A Study of LoRA Fine-Tuning for Text-to-SQL Applications

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## 1 Overview and Task

### 1.1 What is LoRA?

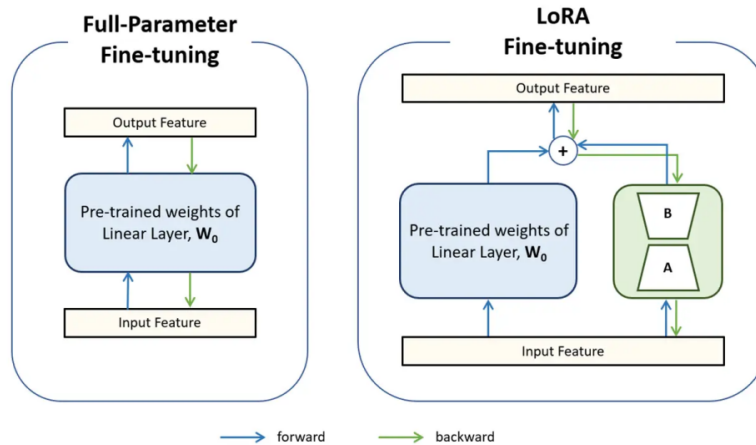


Figure 1: Overview of LoRA.

LoRA is a fine-tuning technique for LLMs. If we want the model to learn a new writing style or customize a general LLM for a specific task, LoRA is very helpful because it avoids training the full model with billions of parameters.

The concept of LoRA is somewhat similar to 'ResNet': we keep the main model, but we add residual components and train them. These components are built from two matrices, A and B. A has the shape  $(d, r)$ , and B has the shape  $(r, d)$ , where  $r$  stands for rank and  $d$  stands for the dimension of the input feature. I will not go into detail because you can read a lot about it on the internet.

### 1.2 What does this project do?

Purpose:

- It helps me learn about LoRA.
- Compare different 'r' hyperparameters.
- Apply the fine-tuned model to a useful case. In my project, the task is: 'Given the context, what is the SQL code for this requirement?' without sending data over the network or using third-party models like DeepSeek, Gemini, or GPT.
- Due to limited computing resources, I treat 'Google Colab' as my private local environment with a T4 GPU.

### 1.3 Resource

- **Base model:** TinyLlama/TinyLlama-1.1B-Chat-v1.0, with approximately 1.1B parameters and 2.2 GB in FP16.
- **Dataset:** b-mc2/sql-create-context, using 10,000 samples: 9,500 for training and 500 for validation.
- **LoRA fine-tuning:** Applied to  $W_q$ ,  $W_k$ ,  $W_v$ , and  $W_o$  projections in all multi-head attention layers.
- **Compute:** NVIDIA T4 GPU with 16 GB VRAM on Google Colab.

## 2 Comparative Research

### 2.1 Training progress

In this project, I chose to adjust 'r' and observe how the results change. The LoRA scaling factor was set to  $\alpha = 2r$ . You can open the notebook to see the details of the other parameters.



Figure 2: Training progress for  $r = 8$  with 2,252,800 trainable parameters.

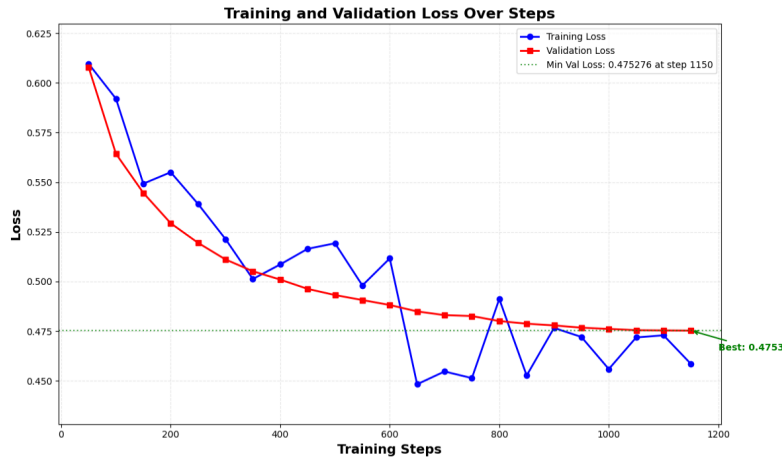


Figure 3: Training progress for  $r = 16$  with 4,505,600 trainable parameters.

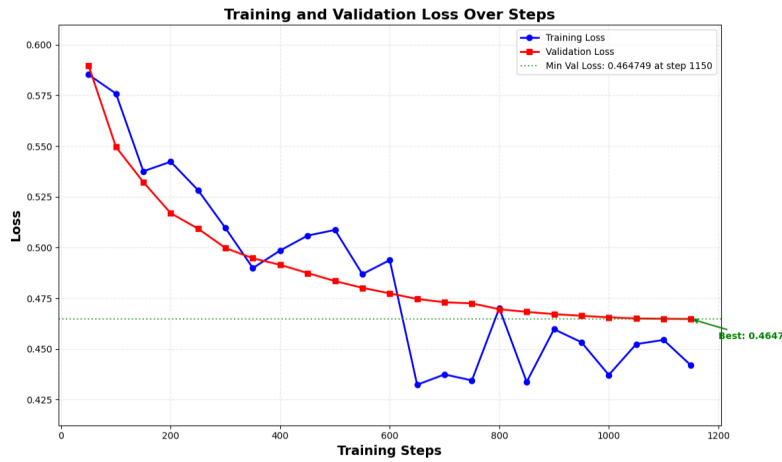


Figure 4: Training progress for  $r = 32$  with 9,011,200 trainable parameters.

Increasing the LoRA rank increases the number of trainable parameters and can improve adaptation capacity. However, a higher rank may overfit the training dataset, while a lower rank may make only a negligible change. Intuitively, from  $r=8$  to  $r=32$ , you can see that a larger rank leads to better train/validation results.

## 2.2 Results

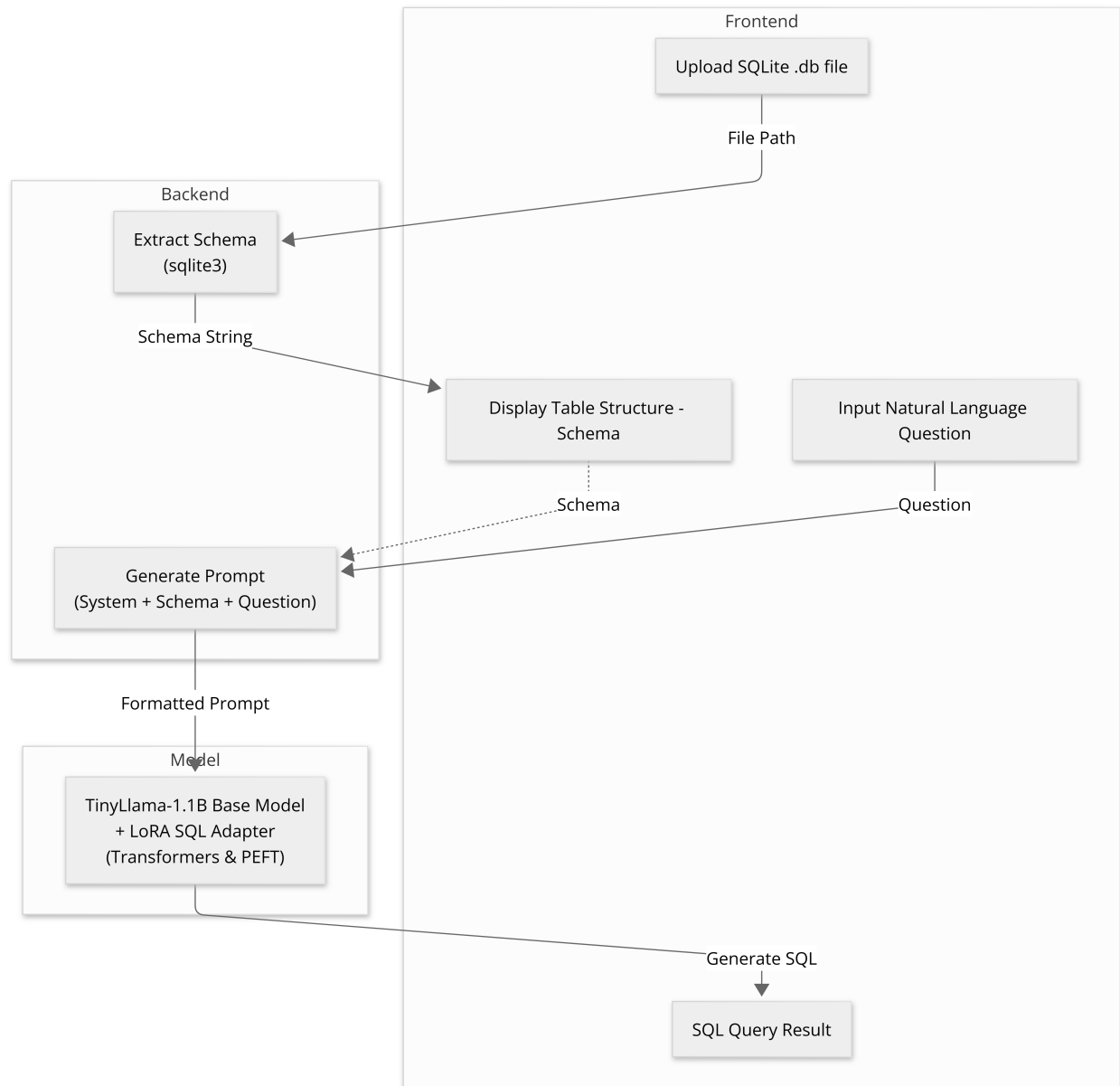
Three test questions are fed into each model. We have four models: the base model and the base model combined with three LoRA adapters of different ranks after training.

| Question                                                                            | Original model                                                                                                                                                                                                                                                                                                                                                                                                                   | R = 8                                                                                                                                             | R = 16                                                                                             | R = 32                                                                                                                                                                          |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| List the names of employees in the Engineering department earning more than 100000. | To answer this question, you can use the following SQL query:<br>SELECT name<br>FROM employees<br>WHERE department = 'Engineering'<br>AND salary > 100000;<br>This query will return a list of names of employees in the Engineering department who have a salary greater than 100,000.                                                                                                                                          | SELECT T1.name FROM employees AS T1 JOIN departments AS T2 ON T1.department = T2.id WHERE T2.name = "Engineering" AND T1.salary > 100000          | SELECT name FROM employees WHERE department = "Engineering" AND salary > 100000                    | SELECT T1.name FROM employees AS T1 JOIN employees AS T2 ON T1.department = T2.department WHERE T2.salary > 100000 AND T1.department = "Engineering"                            |
| What is the total order amount per customer in 2024?                                | To answer the question, you need to use the SUM function to calculate the total order amount per customer in 2024. Here's the SQL query:<br>SELECT SUM(amount) AS total_order_amount_per_customer<br>FROM orders<br>WHERE year(order_date) = 2024<br>GROUP BY customer_id;<br>This query will return a single row with the total order amount per customer in 2024.                                                              | SELECT SUM(amount) FROM orders WHERE customer_id = 2024 AND order_date > 2024                                                                     | SELECT SUM(amount) FROM orders WHERE customer_id = 2024 GROUP BY order_id                          | SELECT SUM(amount), T1.customer_id, T2.order_date FROM orders AS T1 JOIN customers AS T2 ON T1.customer_id = T2.customer_id WHERE T2.customer_id = 2024 GROUP BY T1.customer_id |
| Show the top 5 highest rated horror movies released after 2015.                     | To show the top 5 highest rated horror movies released after 2015, you can use the following SQL query:<br>SELECT title, year, rating, COUNT(*) AS total_ratings<br>FROM movies<br>WHERE year > 2015<br>GROUP BY title, year<br>ORDER BY total_ratings DESC<br>LIMIT 5;<br>This query will return the top 5 highest rated horror movies released after 2015, ordered by total ratings descending. The GROUP BY clause groups the | SELECT T1.title, T1.year, T1.rating, T2.genre FROM movies AS T1 JOIN movies AS T2 ON T1.year > 2015 WHERE T1.genre = 'Horror' AND T1.rating > 9.0 | SELECT title, COUNT(*) FROM movies WHERE year > 2015 GROUP BY title ORDER BY COUNT(*) DESC LIMIT 5 | SELECT T1.title, T1.year, T1.rating, T2.genre FROM movies AS T1 JOIN movies AS T2 ON T1.year > 2015 WHERE T1.genre = 'Horror' ORDER BY T1.rating DESC LIMIT 5                   |

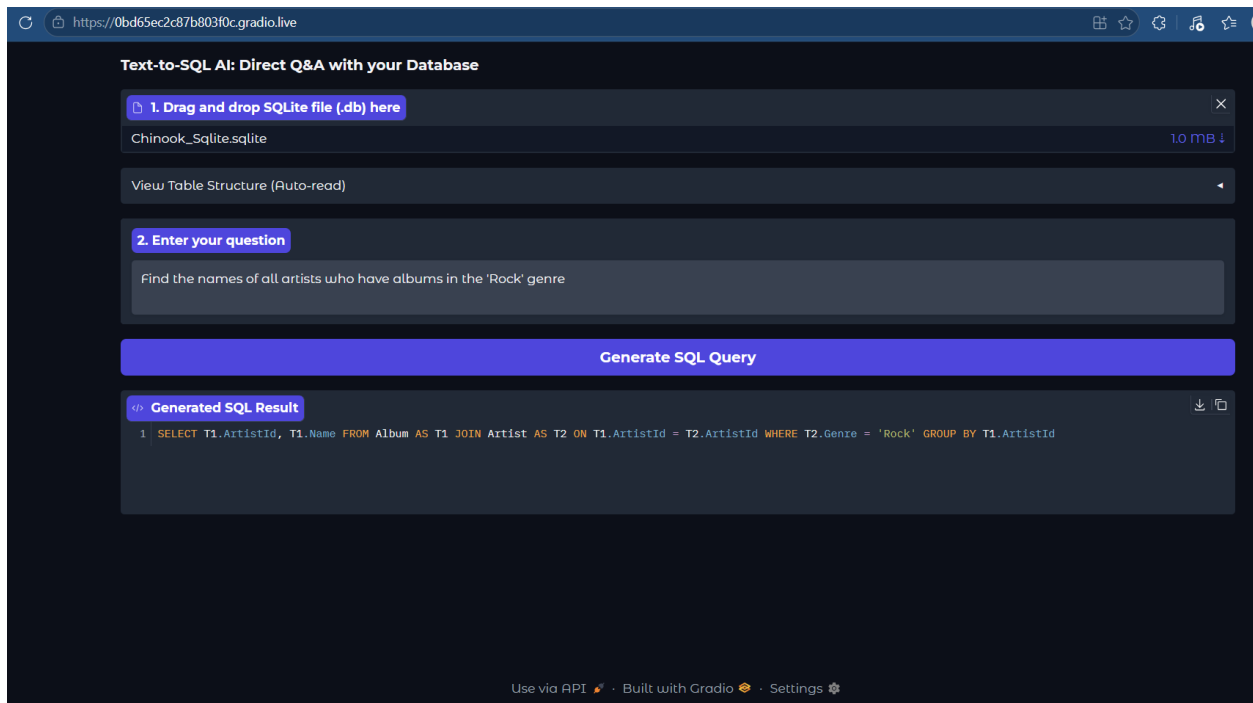
Table 1: Training steps = 1188

Basically, after LoRA fine-tuning, the answers are shown as desired: they are short and concise. However, if you check their meaning and suitability, they are wrong or partially wrong. The base model gives 2 out of 3 wrong answers. This means that if we want better answers for most questions, we need to train the full model again with more abundant SQL samples.

### 3 System Design and Implementation



The simple system is designed as shown in the diagram above. The framework I use is 'Gradio', which helps create UIs quickly for AI/ML projects. Here is an example of a single question. Do not forget to upload the database file first (.sqlite or .db), and then see how the model responds.



## 4 Reference and source code

Base reference: [https://drive.google.com/file/d/18\\_n8VYa287Hf2s0PAKE\\_G9ok8ptyvspn/view](https://drive.google.com/file/d/18_n8VYa287Hf2s0PAKE_G9ok8ptyvspn/view)

Modified reference: <https://colab.research.google.com/drive/1sHZwHrD0tqihXqHB8KeQfy44PF7C17i0?usp=sharing>

System notebook: <https://colab.research.google.com/drive/1kwoAjXUC0kmUNOjtyV7ghHWh2JJv5WgB?usp=sharing>

Full source code and additional files <https://github.com/>